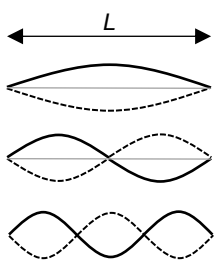


Qns	Answer	Marks
9(a)(i)	[source of force] wire experiences magnetic force directed normal to both current and magnetic field	1
	[property of vibration] alternating current so direction of force oscillates between vertically upwards and downwards	1
9(a)(ii)1.	$v = f\lambda = (50)(2 \times (40 \times 10^{-2})) = 40 \text{ m s}^{-1}$	1
9(a)(ii)2.	wave of same type, same frequency, same speed, (same amplitude)	1
	are reflected off fixed ends X and P and travel towards each other in opposite directions	1
	waves superpose along length XP	1
	speed is that (of energy transfer) of individual/constituent incident / reflected wave in wire	1
9(a)(ii)3.	natural frequency of wire changes and no longer matches driving frequency from alternating current	1
	wire no longer in resonance so amplitude decrease	1
9(a)(ii)4.	$L = \lambda_1$ $f_1 = \frac{v}{\lambda_1} = \frac{40}{40 \times 10^{-2}} = 100 \text{ Hz}$ $L = \frac{3}{2} \lambda_2$ $f_2 = \frac{v}{\lambda_2} = \frac{40}{\left(\frac{2}{3}\right) 40 \times 10^{-2}} = 150 \text{ Hz}$ 	1
9(b)(i)	$A = A_0 \exp(-\lambda t)$ $\ln A = \ln A_0 - \lambda t$ The gradient of the $\ln (A / \text{s}^{-1})$ against t graph will give the value for $-\lambda$. $\ln (A / \text{s}^{-1})$ in 1990 = 16.80; $\ln (A / \text{s}^{-1})$ in 1997 = 16.20 $\text{gradient} = \frac{y_1 - y_2}{x_1 - x_2} = \frac{16.20 - 16.80}{7} = -0.0857 \text{ y}^{-1}$ $\lambda = 2.72 \times 10^{-9} \text{ s}^{-1}$	1
		1
		1
9(b)(ii)	$T_{\frac{1}{2}} = \frac{\ln 2}{\lambda}$ $T_{\frac{1}{2}} = \frac{\ln 2}{2.72 \times 10^{-9}} = 2.55 \times 10^8 \text{ s}$ $= 8.1 \text{ yr}$	1
		1

Qns	Answer	Marks
9(b)(iii)	At year 1997, $\ln(A / \text{s}^{-1}) = 16.20$ $A = e^{16.20} = 1.10 \times 10^7 \text{ s}^{-1}$.	1
	The number of nuclei which ought to be present is $N_1 = \frac{A}{\lambda} = \frac{1.10 \times 10^7}{2.718 \times 10^{-9}}$ $= 4.00 \times 10^{15}$	1
9(b)(iv)	The number of nuclei left after the theft is $N_2 = \frac{A}{\lambda} = \frac{e^{15.92}}{2.718 \times 10^{-9}} = 3.02 \times 10^{15}$	1
	The number of nuclei stolen is given by $4.00 \times 10^{15} - 3.02 \times 10^{15} = 0.975 \times 10^{15}$	1
9(b)(v)	gradient corresponds to the (magnitude of) decay constant	
	decay constant is <u>not dependent on the number of radioactive nuclei</u>	1
9(b)(v)	but is <u>characteristic/unique</u> to a particular nuclide	1